

AdaptiveThink-MoE: Execution Action Plan (V1)

Objective: Build a practical Mixture-of-Experts style system using a shared Qwen base model, specialized LoRA experts, and a routing layer. Focus on delivering a working research project that can be benchmarked and shared publicly.

Phase 0 – Foundation (Current Sprint)

- Create repository structure.
- Set up Python environment, Docker/dev container (optional).
- Install transformers, peft, trl, datasets, accelerate, bitsandbytes, sentence-transformers, wandb.
- Load Qwen2.5-0.5B-Instruct and verify inference.
- Commit baseline project to GitHub.

Phase 1 – Code Expert

- Download MBPP dataset.
- Inspect dataset schema.
- Convert samples into instruction-response format.
- Create a 500-sample subset for initial experiments.
- Train LoRA adapter on the subset.
- Validate generation quality.
- Scale training to full dataset.

Phase 2 – Math Expert

- Download GSM8K dataset.
- Format data into instruction-response pairs.
- Train a separate LoRA adapter.
- Evaluate on GSM8K test samples.

Phase 3 – General Expert

- Download Alpaca-style instruction dataset.
- Train General LoRA adapter.
- Evaluate general assistant behavior.

Phase 4 – Router

- Use all-MiniLM-L6-v2 embeddings.
- Create expert-labeled prompts.
- Build routing mechanism.

- Route prompts to Code, Math, or General expert.

Phase 5 – Inference Pipeline

- Load base model once.
- Load expert adapter dynamically.
- Generate responses through selected expert.
- Create CLI interface.

Phase 6 – Benchmarking

- Code Expert → HumanEval.
- Math Expert → GSM8K.
- General Expert → instruction-following evaluation.
- Compare against base Qwen model.

Phase 7 – Research Extensions

- Top-K expert routing.
- Difficulty-aware routing.
- Adaptive compute allocation.
- Publish findings and benchmarks.

Recommended Weekly Timeline

- Week 1: Foundation + Dataset Pipeline
- Week 2: Code Expert Training
- Week 3: Math Expert Training
- Week 4: General Expert Training
- Week 5: Router Development
- Week 6: Inference Pipeline
- Week 7: Benchmarking
- Week 8: Research Extensions + LinkedIn Publication